

LESSON 9-4 • HOMEWORK PACKET • SHEET 1 OF 2**Substitute and Solve**

Maryland Standard 6.EE.C.9 • Learning target – I can substitute a value into an equation to find the matching output.

REMEMBER

- To find the output, **substitute** the input for x .
- Then use **order of operations**: multiply first, then add or subtract.

WORKED EXAMPLEUse $y = 6x + 4$ to find y when $x = 5$.**Step 1** Substitute: $y = 6 \times 5 + 4$ **Step 2** Multiply, then add: $30 + 4 = 34$ **Answer $y = 34$, so $(5, 34)$** **PRACTICE***show your work • label units*

1. Substitute to find
- y
- .

a. $y = 4x + 7$, $x = 3$ _____

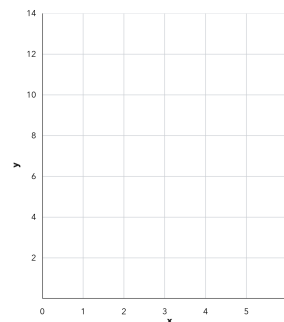
b. $y = 9x$, $x = 8$ _____

c. $y = 3x - 2$, $x = 10$ _____

d. $y = x + 15$, $x = 27$ _____

2. Complete the table for
- $y = 2x + 3$
- . Then plot the points and connect them.

x	0	1	2	4	5
y					



- 3.
- Water Tank.**
- A tank has 20 gallons. It fills at 15 gallons each minute, so
- $g = 15m + 20$
- .

a. Gallons after 6 minutes: _____ gal

b. Gallons after 10 minutes: _____ gal

- 4.
- Bike Rental.**
- A bike rental costs
- $c = 7h + 5$
- dollars for
- h
- hours. Find the cost of renting for 3 hours. Then find the cost for 8 hours.

Costs _____

- 5.
- On the Line?**
- Substitute
- x
- to decide whether each point is on the graph of
- $y = 5x + 2$
- .

a. $(4, 22)$: $5 \times 4 + 2 =$ _____ On the line? ☐ Yes ☐ No

b. $(6, 30)$: $5 \times 6 + 2 =$ _____ On the line? ☐ Yes ☐ No

- 6.
- Find the Error.**
- For
- $y = 4x + 3$
- with
- $x = 5$
- , Dev added first:
- $4 + 3 = 7$
- , and
- $7 \times 5 = 35$
- . Explain his mistake and find
- y
- .

CHECK YOURSELF☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** Pick a rule like $y = 2x + 1$. Ask someone at home for an input and find the output together.

LESSON 9-4 • HOMEWORK PACKET • SHEET 2 OF 2**Working Backward to a Goal**

Maryland Standard 6.EE.C.9 • Learning target – I can work backward from a goal to find the input an equation needs.

REMEMBER

- Know the goal (y)? Substitute it, then solve for x .
- Undo in **reverse order**: subtract the fee first, then divide by the rate.
- In a counting problem, a decimal answer may need to be **rounded up**.

WORKED EXAMPLEA rental costs $y = 4x + 10$ for x hours. How many hours cost \$46?**Step 1** Subtract 10: $46 - 10 = 36$ **Step 2** Divide by 4: $36 \div 4 = 9$ **Answer 9 hours****PRACTICE***show your work • label units*

1. Solve for
- x
- .

a. $7x = 63$ _____

b. $8x = 104$ _____

c. $3x + 5 = 32$ _____

d. $6x + 12 = 60$ _____

- 2.
- Trip Fund.**
- A class has \$45 and earns \$15 a week:
- $y = 15x + 45$
- . How many weeks until it has \$210?

Answer _____ weeks

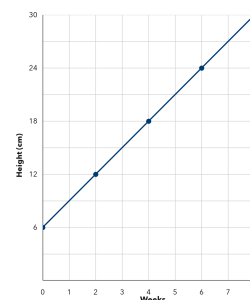
- 3.
- Gift Fund.**
- A club has \$30 in its gift fund. Members add \$12 each week, so
- $y = 12x + 30$
- . How many weeks until the fund has \$150?

Answer _____ weeks

- 4.
- Plant Growth.**
- A plant is 6 cm tall and grows 3 cm a week:
- $h = 3w + 6$
- .

a. Use the graph. When is it 18 cm? _____

b. Use the equation. When is it 33 cm? _____



- 5.
- Round Up.**
- A box holds 12 cookies. Ana needs 150 cookies, so she says 12.5 boxes. How many boxes should she buy? Explain.

TEST PRACTICE*choose the best answer*

6. A rental costs
- $y = 8x + 12$
- dollars for
- x
- hours. Max has \$60. How many hours can he rent?

A 5 hours**B** 6 hours**C** 7.5 hours**D** 9 hours

7. Mia solved
- $5x + 20 = 85$
- by dividing 85 by 5 first. What is the correct value of
- x
- ?

A $x = 17$ **B** $x = 21$ **C** $x = 13$ **D** $x = 65$ **CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about #_____